
Lab 1

HW / SW Co-design of audio spectrum

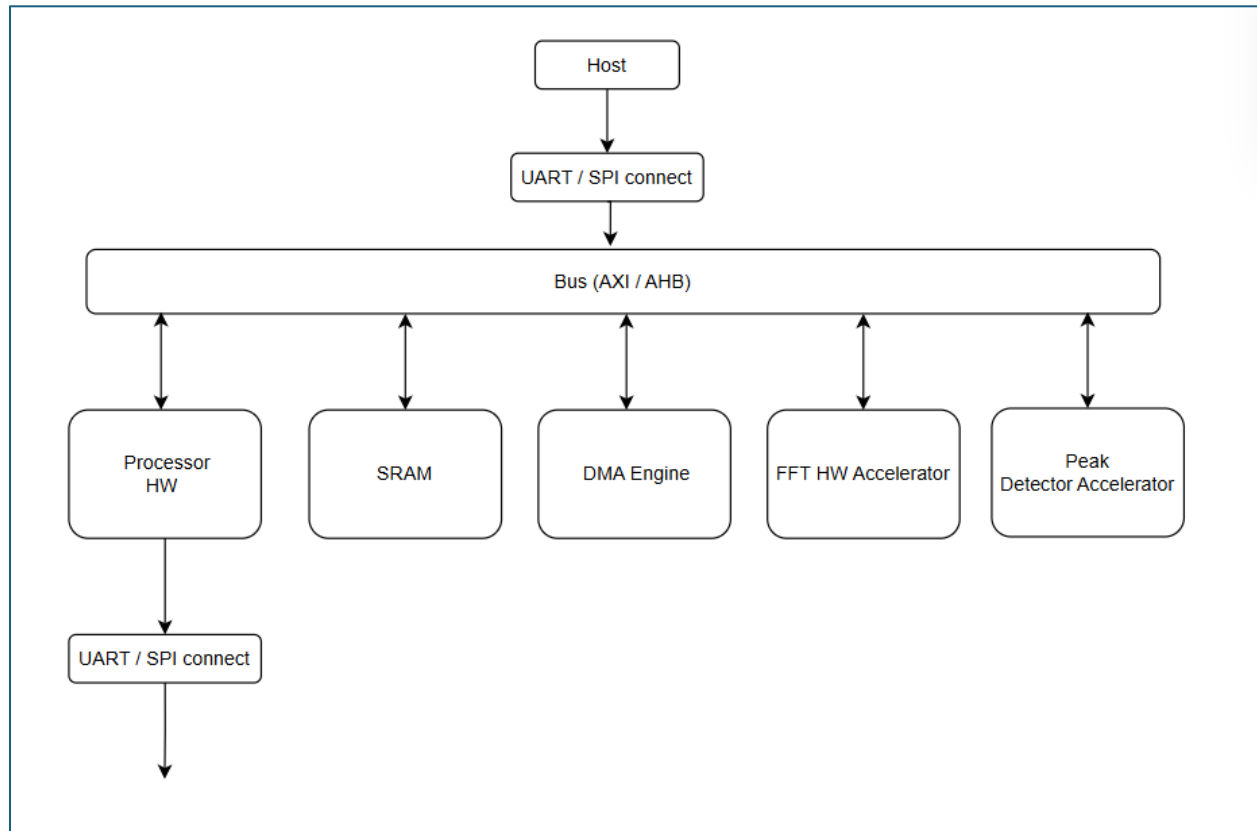
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1. SoC Arch.



2. HW / SW

1. Processor

- Configure peripherals
- Control the whole application
- Start FFT accelerator
- Read FFT results
- Convert bin index into frequency
- Send final answer to PC

HW. It's pure repetitive data movement.

3. Memory (SRAM)

Stores:

- Input samples
- FFT output
- Magnitude values

physical memory used to store samples and processing results → **HW**

4. DMA (optional but recommended)

automatically transfers samples from UART to memory.

- Faster
- CPU free for other work

Data transfer happens frequently and high performance is required → **HW**

5. FFT Hardware Accelerator

computationally intensive operation. → **HW**

6. Peak Detector Hardware

Scans FFT magnitudes and Finds peak magnitude

It is a simple calculation that may change easily if the FFT size or sampling rate changes, so software is more flexible → **SW**

3. Psuedo code

```
Start
Initialize UART/SPI
Initialize DMA
Initialize FFT Accelerator
Receive audio samples from Host
Transfer samples to SRAM using DMA
Start FFT Accelerator
Wait until FFT is finished
Read peak frequency bin
Calculate frequency
Send frequency back to Host through UART/SPI
End
```